

JAVA LAB RECORD

Experiment No.	06
Date	23/09/26
Title	Implementation of Vector Methods in Java

Objective / Aim

To study and implement commonly used methods of the Java Vector class, including adding, accessing, updating, removing, searching, and checking the size and capacity of a Vector.

System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Keyboard and mouse
- Monitor/display

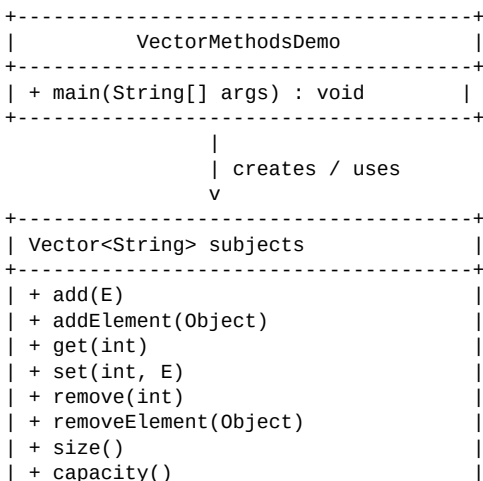
Software Requirements

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK): 17 or later
- IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program.
2. Import the Vector class from java.util.
3. Create a Vector named subjects to store String values.
4. Add subject names using add(E e).
5. Add another subject using addElement(Object obj).
6. Retrieve an element at a specified index using get(index).
7. Replace an element using set(index, element).
8. Remove an element using its index with remove(index).
9. Remove a specific element using removeElement(Object obj).
10. Find the number of elements using size().
11. Find the current storage capacity using capacity().
12. Check whether particular elements exist using contains(Object).
13. Display all results.
14. Stop the program.

UML Class Diagram



```
| + contains(Object) |
+-----+
```

Source Code

```
import java.util.Vector;

class VectorMethodsDemo {

    public static void main(String[] args) {

        // Creating a Vector
        Vector<String> subjects = new Vector<>();

        // add(E e)
        subjects.add("Java");
        subjects.add("Python");
        subjects.add("C++");

        System.out.println("After add(): " + subjects);

        // addElement(Object obj)
        subjects.addElement("DBMS");
        System.out.println("After addElement(): " + subjects);

        // get(int index)
        System.out.println("Element at index 1: " + subjects.get(1));

        // set(int index, E element)
        subjects.set(2, "C");
        System.out.println("After set(): " + subjects);

        // remove(int index)
        subjects.remove(1);
        System.out.println("After remove(index): " + subjects);

        // removeElement(Object obj)
        subjects.removeElement("DBMS");
        System.out.println("After removeElement(): " + subjects);

        // size()
        System.out.println("Size of Vector: " + subjects.size());

        // capacity()
        System.out.println("Capacity of Vector: " + subjects.capacity());

        // contains(Object o)
        System.out.println("Contains Java? " + subjects.contains("Java"));
        System.out.println("Contains Python? " + subjects.contains("Python"));
    }
}
```

Completion and Execution Commands

1. Save the program as:
VectorMethodsDemo.java
2. Open Command Prompt / Terminal and move to the folder containing the Java file.
3. Compile the program:
javac VectorMethodsDemo.java
4. Execute the program:
java VectorMethodsDemo
5. The results of all Vector operations are displayed on the console.

Output

```
After add(): [Java, Python, C++]
After addElement(): [Java, Python, C++, DBMS]
Element at index 1: Python
After set(): [Java, Python, C, DBMS]
After remove(index): [Java, C, DBMS]
After removeElement(): [Java, C]
```

Size of Vector: 2
Capacity of Vector: 10
Contains Java? true
Contains Python? false

Result

Thus, the Java program was successfully implemented and executed to demonstrate the commonly used methods of the Vector class.